

CS-1002 Programming fundamentals

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Do-while loop examples

- Example to sum all the numbers entered by user and stop process if user enter –ve value

```
Terminal Help demo.cpp - C++  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL  
PS D:\Desktop\C++> cd "d:\Desktop\C++\" ;  
enter a number:1  
enter a number:10  
enter a number:2  
enter a number:20  
enter a number:-5  
sum=33  
PS D:\Desktop\C++> 
```

**Sample
Input/
Output**

Coding Exercise (do-while loop)

```
int main()
{
    int number, sum = 0;
    do {
        cout << "Enter a number: ";
        cin >> number;
        sum = sum + number;
    } while (number >= 0);
    cout << "The sum is: " << sum;
    return 0;
}
```

Enter a number: 1
 Enter a number: 10
 Enter a number: 2
 Enter a number: 20
 Enter a number: -5
 The sum is: 33

**Invalid
 Solution
 Because its
 adding -ve
 value too so
 the sum will
 not be 33**

```
demo.cpp - C++ - Visual Studio Code
while_Loop.cpp demo.cpp X whileloop.cpp
cpp > main()
#include<iostream>
using namespace std;

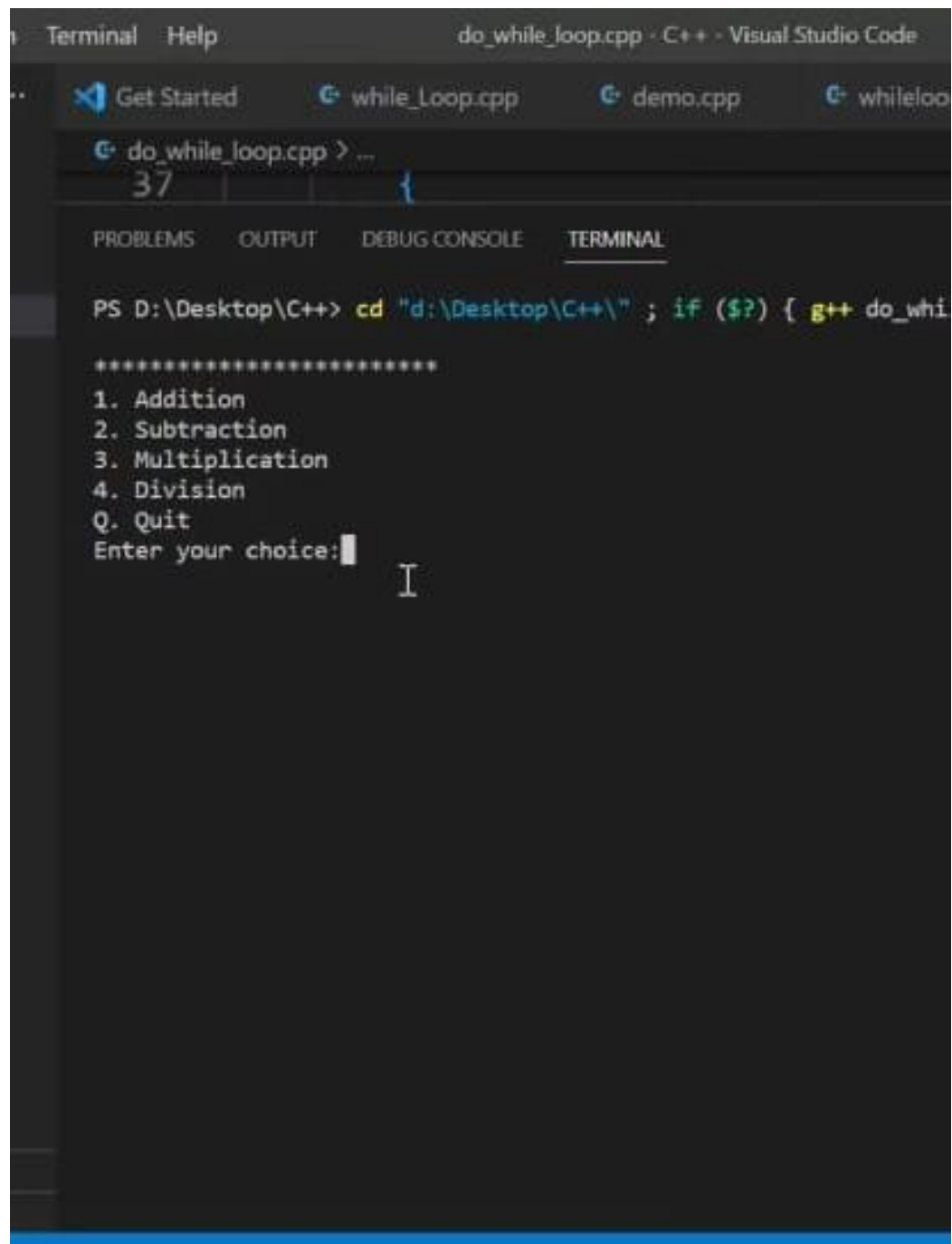
int main()
{
    int number=0,sum=0;
    do
    {
        sum += number;
        cout<<"enter a number:";
        cin>>number;
    } while (number>=0);

    cout<<"sum="<<sum<<endl;
    return 0;
}
```

**Correct
Solution: to
sum before
taking input**

- Another Example

Sample Input/Output

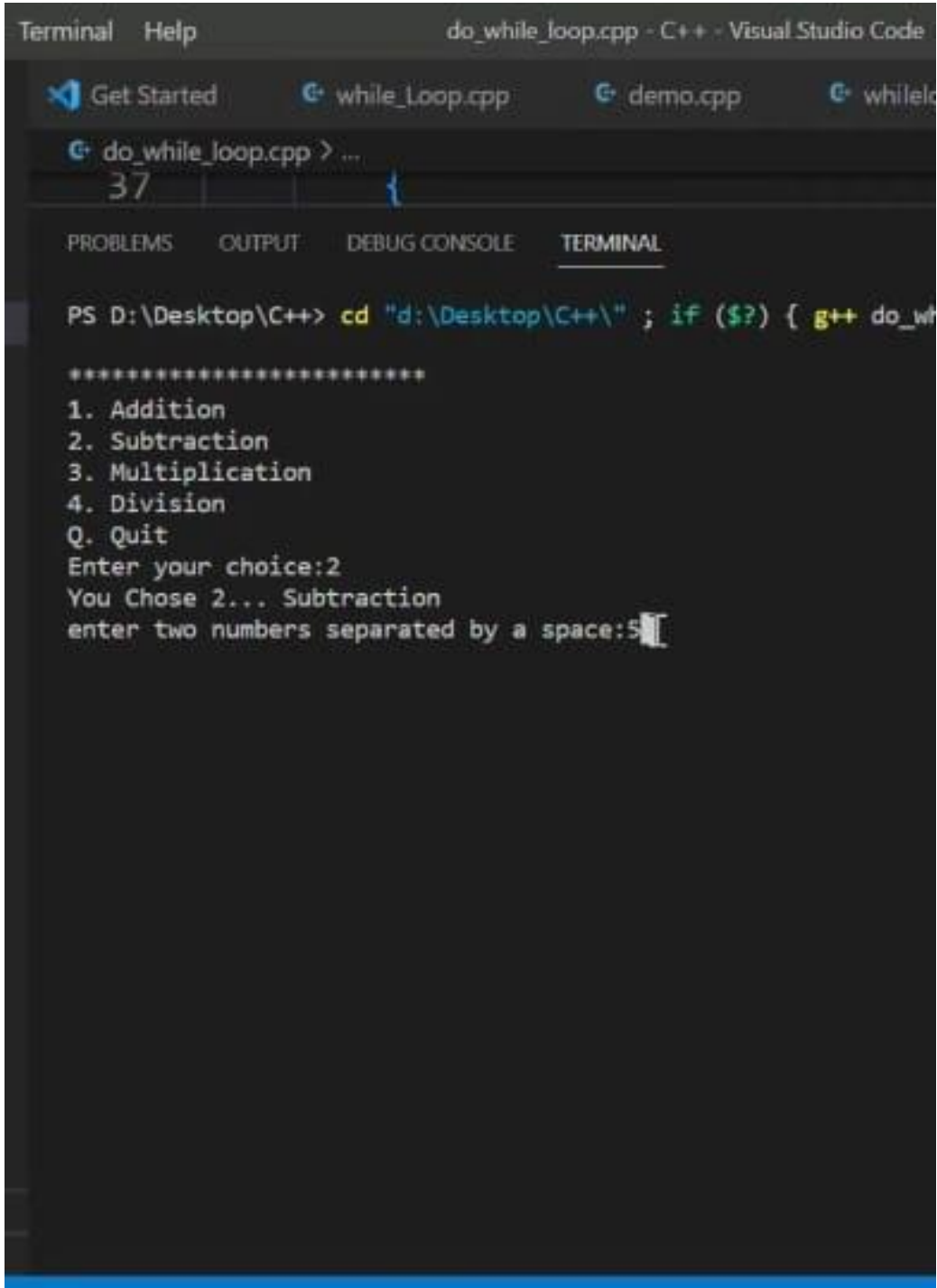


The image shows a Visual Studio Code window with a C++ project. The top bar indicates the file is `do_while_loop.cpp` and the language is C++. The Explorer sidebar on the left shows the file structure with `do_while_loop.cpp` selected. The main editor area displays the code for `do_while_loop.cpp`, showing line 37 with a closing curly brace `}`. The TERMINAL panel at the bottom shows the command prompt output. The command executed is `PS D:\Desktop\C++> cd "d:\Desktop\C++\" ; if ($?) { g++ do_while_loop.cpp -o do_while_loop.exe ; } && .\do_while_loop.exe`. The program output lists four options: 1. Addition, 2. Subtraction, 3. Multiplication, and 4. Division, followed by "Q. Quit" and a prompt "Enter your choice:". The cursor is positioned at the end of the prompt, ready for user input.

```
do_while_loop.cpp > ...
37 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS D:\Desktop\C++> cd "d:\Desktop\C++\" ; if ($?) { g++ do_while_loop.cpp -o do_while_loop.exe ; } && .\do_while_loop.exe
*****
1. Addition
2. Subtraction
3. Multiplication
4. Division
Q. Quit
Enter your choice: |
```



The image shows a Visual Studio Code window with the file `do_while_loop.cpp` open. The editor shows line 37 with a closing curly brace `}`. The terminal window is active, displaying the command `cd "d:\Desktop\C++\" ; if ($?) { g++ do_w` and the output of a C++ program. The program prompts the user to choose an operation from a list: 1. Addition, 2. Subtraction, 3. Multiplication, 4. Division, and Q. Quit. The user has entered '2', and the program has responded 'You Chose 2... Subtraction'. It then prompts the user to 'enter two numbers separated by a space:'. The cursor is positioned at the end of the prompt.

```
Terminal Help do_while_loop.cpp - C++ Visual Studio Code

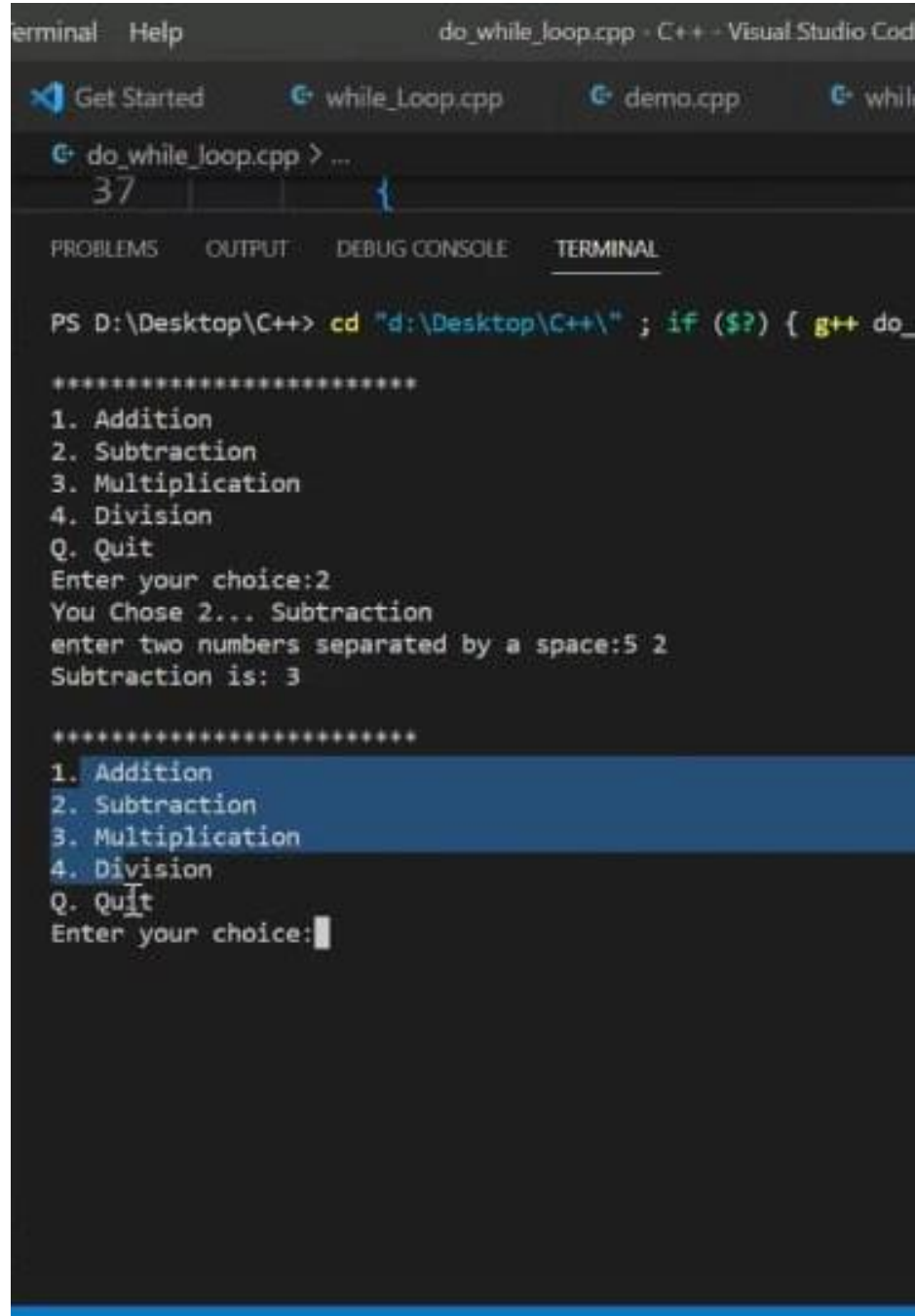
Get Started while_loop.cpp demo.cpp whilel

do_while_loop.cpp > ...
37 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS D:\Desktop\C++> cd "d:\Desktop\C++\" ; if ($?) { g++ do_w

*****
1. Addition
2. Subtraction
3. Multiplication
4. Division
Q. Quit
Enter your choice:2
You Chose 2... Subtraction
enter two numbers separated by a space:5
```



```
terminal Help do_while_loop.cpp C++ - Visual Studio Cod

Get Started while_loop.cpp demo.cpp while

do_while_loop.cpp > ...
37 {

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

PS D:\Desktop\C++> cd "d:\Desktop\C++\" ; if ($?) { g++ do_

*****
1. Addition
2. Subtraction
3. Multiplication
4. Division
Q. Quit
Enter your choice:2
You Chose 2... Subtraction
enter two numbers separated by a space:5 2
Subtraction is: 3

*****
1. Addition
2. Subtraction
3. Multiplication
4. Division
Q. Quit
Enter your choice:
```

```
> cd "d:\Desktop\C++\" ; if ($?) { g++ do_

*****
1. Addition
2. Subtraction
3. Multiplication
4. Division
Q. Quit
Enter your choice:7
Wrong choice... Try again!!

*****
1. Addition
2. Subtraction
3. Multiplication
4. Division  I
Q. Quit
Enter your choice:█
```

Code

```
calculator.cpp > main()
1  #include<iostream>
2  using namespace std;
3  int main()
4  {
5      char choice;
6      do{
7          cout<<"\n*****"<<endl;
8          cout<<"1. Addition"<<endl;
9          cout<<"2. Subtraction"<<endl;
10         cout<<"3. Multiplication"<<endl;
11         cout<<"4. Division"<<endl;
12         cout<<"Q. Quit"<<endl;
13         cout<<"enter your choice:";
14         cin>>choice;
15
16
17
18
19
20
```

```
al Help • calculator.cpp - C++ - Visual Studio Code
et Started while_loop.cpp demo.cpp whileloop.cpp do_while_loop.cpp calculate
calculator.cpp > main()
34     else if(choice == '3')
35     {
36         cout<<"You Chose 3... Multiplication"<<endl;
37         cout<<"enter two numbers separated by a space:"
38         cin>>a>>b;
39         mul=a*b;
40         cout<<"Multiplication is: "<<mul<<endl;
41     }
42     else if(choice == '4')
43     {
44         cout<<"You Chose 4... Division"<<endl;
45         cout<<"enter two numbers separated by a space:"
46         cin>>a>>b;
47         div=a/b;
48         cout<<"Division is: "<<div<<endl;
49     }
```

```
calculator.cpp > main()
45     cout<<"enter two numbers separated by a space:";
46     cin>>a>>b;
47     div=a/b;
48     cout<<"Division is: "<<div<<endl;
49 }
50 else if(choice == 'q' || choice == 'Q')
51 {
52     cout<<"Good Bye..."<<endl;
53 }
54 else{
55     cout<<"Wrong Choice... Try again"<<endl;
56 }
57
58
59
```

```
ator.cpp > main()

    cout<<"enter two numbers separated by a space:";
    cin>>a>>b;
    div=a/b;
    cout<<"Division is: "<<div<<endl;
}
else if(choice == 'q' || choice == 'Q')
{
    cout<<"Good Bye..."<<endl;
}
else{
    cout<<"Wrong Choice... Try again"<<endl;
}
}while(choice != 'q' && choice != 'Q');
}
```